Jianhao Zhang, Ph.D.

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Research Interests

My research characterizes and quantifies how clouds respond to changes in aerosol loading and meteorological conditions driven by human activity and natural climate variability. Building on this, I seek to understand how these cloud responses shape the increasing Earth's energy imbalance and constrain the viability and risks of proposed climate interventions. To address these multiscale questions, I leverage satellite and field observations, atmospheric and Earth system modeling, and data-driven approaches to extract physical understanding of the processes linking clouds to a changing climate.

Professional Appointment

May 2026 – present — **Research Scientist III,** NOAA Chemical Sciences Laboratory & CIRES at the University of Colorado Boulder

May 2023 – Apr 2026 — **Research Scientist II,** NOAA Chemical Sciences Laboratory & CIRES at the University of Colorado Boulder

Sept. 2021 – Apr. 2023 — **Research Scientist I,** NOAA Chemical Sciences Laboratory & CIRES at the University of Colorado Boulder

Sept. 2020 – Aug. 2021 — **NRC Postdoc fellowship** hosted by Graham Feingold at NOAA Chemical Sciences Laboratory

Education

2020 — **Ph.D., University of Miami, Miami, FL, USA** in Meteorology & Physical Oceanography.
Dissertation title: *The interactions between light-absorbing smoke and marine boundary layer clouds over the remote southeast Atlantic.*
Committee: *Paquita Zuidema (Chair), Brian Mapes, Brian Soden, Cassandra Gaston, David Turner & Takanobu Yamaguchi*

2014 — **B.S., Florida State University, Tallahassee, FL, USA** in Meteorology.
Magna cum laude; Minor in Mathematics.

Funded Grants

2026-ongoing — **Co-I.:** Assessing the detectability of small-scale stratospheric aerosol injections (**DOD** ESTCP Natural Hazards Project NH26-9821)

2025-2026 — **P. I.:** A Framework for Assessing SRM Detectability for Informed Decision-Making (**Reflective**, $162,768)

2023-2026 — **Co-I.:** Aerosol-Cloud Interactions Centered on MAGIC: Insights from Measurements and Lagrangian Large Eddy Simulation (**DOE** Atmospheric System Research, $670,445)

2023-2024 — **P.I. for CIRES:** Exploring Aerosol-Cloud Interactions in Geophysical Variable Spaces using NASA-ACTIVATE Observations (**NASA** Earth Venture Suborbital-3, $350,000)

2020-2023 — **Co-I.:** Evaluating Biases in Aerosol-Cloud Interaction Metrics using ARM Data and Models (**DOE** Atmospheric System Research, $510,472)

2020-2021 — **P. I.:** National Research Council Fellowship Award (**National Academies of Sciences, Engineering, Medicine**, $62,000)

Awards, Fellowships, & Honors

- **NOAA Boulder Outreach Gold Star Award, 2025**
- **Reflective Fellowship, 2025**, Reflective Inc.
- **National Research Council Research Associateship Fellowship, 2020**, National Academies of Sciences, Engineering, and Medicine
- **NASA's Group Achievement Awards** (ACTIVATE Team, 2023; ORACLES Team, 2019)
- **Meteorology & Physical Oceanography Program Best Ph.D. Dissertation, 2020**
- **University of Miami Graduate Fellowship, 2014** (highest award of the graduate school)

Teaching, Mentoring, & Outreach

Teaching
- **The Program for Exploratory Studies** as *Guest lecturer*, CU Boulder (Apr 2026)
- **Aerosol-Cloud-Climate Interactions** as *Guest lecturer*, TU Delft (June 2024)
- **Introduction to Weather and Climate** as *Teaching Assistant*, University of Miami (2015)
- **AP calculus and undergrad statistics** as *Math Tutor*, Miami (2015-2020)

Mentoring
- **Tyler Tatro**, Ph.D. student at University of Miami (2022-present)
- **Alexander J. Thompson**, Research Scientist at NOAA/PSL (2024-2025)
- **Danyan Leng**, Ph.D. student at University of Colorado Boulder (2025-2026)
- **Daisy Kerr**, NOAA Hollings Scholar, Physics undergrad at UC Berkeley (Summer 2026)
- Dialogic Skills Workshop Certificate, University of Colorado Boulder, 2025

Outreach
- **CIRES Speaker**, CIRES Science Pathways Program (2025-)
- **Scientist Panel**, Environmental studies class at Boulder High School (2025)
- **Panelist**, CIRES Graduate Student Workshop (2025)
- **Invited Scientist**, Boulder Valley Rotary Club Weekly Discussion (2025)
- **Science vetting**, Children's book *UP, UP HIGH*
- **Evaluator**, Research Experience for Community College Students Symposium 2024; GLOBE International Virtual Science Symposium 2022; Climate Literacy and Energy Awareness Network 2022

Leadership & Service

Reviewer
- Department of Energy (DOE) Atmospheric System Research (ASR)
- University of Colorado AB Nexus Program
- NOAA Hollings Undergraduate Scholarship
- Peer-reviewer for 17 major academic journals

Committee
- AGU Dewan Scholarship (2026-present)
- CIRES Members' Council (2025-present)
- CIRES Mentoring Program (2024-present, Chair)
- NOAA OAR subject matter expert in *Satellite Data* (2022-present)
- University of Miami RSMAS Student Seminar Committee (2016)

Convener
- **AMS's** 17th Conference on Atmospheric Radiation, 2026, Madison, WI
- **AMS 2026**, *Advances in Assessing Aerosol Impacts on Clouds Through AI and Advanced Statistical Techniques*, Houston, TX
- **AGU 2024**, *Advances in Cloud and Precipitation Processes: Integrating Observations, Modeling, and Theory*, Washington D.C.
- **AMS's** 16th Conference on Cloud Physics, 2022, Madison, WI

Discussion Lead
- 2025 Gordon Research Seminar on Radiation and Climate, Lewiston, ME
- University of Washington Inaugural MCB Workshop, 2025, Leavenworth, WA

Field campaign
- NASA ORACLES-2016, Swakopmund, Namibia, Sept. 2016
- NASA ORACLES-2017, São Tomé, São Tomé and Príncipe, Aug. 2017

Invited Presentations

May 2026 — **Libera Science Team Meeting, Berkeley, CA,** *Earth's East-West Albedo Symmetry.*

Mar. 2026 — **NOAA Chemical Sciences Laboratory Seminar, Boulder, CO,** *A cloudy story of Earth's radiation budget.*

May 2025 — **AGU AS Early-Career Seminar, Online,** *Global climate and air quality implications of regional emission shift.*

Apr. 2025 — **Boulder Valley Rotary Club Weekly Meeting, Boulder, CO,** *How to measure the shade of clouds that are no longer there.*

- **Climate Dynamics & Impacts, Vecchi/Soden Joint Group Meeting, Princeton University and University of Miami, Online,** *Large radiative forcing from the 2020 shipping fuel regulation is hard to detect: Implications for Marine Cloud Brightening.*
- **University of Washington Inaugural MCB Program Workshop, Leavenworth, WA, Session co-lead,** *Identifying conditions amenable to cloud brightening/MCB.*
- **NOAA Science Seminar Series, Online,** *AI Applications in Earth System and Climate Science: Aerosols and Air Quality.*

Nov. 2024 — **NOAA Advancing Innovative Research Seminar Series, Online,** *Large radiative forcing from the 2020 shipping fuel regulation is hard to detect.*

Jun. 2024 — **TU Delft, Delft, Netherlands,** *On the viability of Marine Cloud Brightening: Albedo susceptibility, cloud adjustment, and detectability.*

Oct. 2023 — **Brookhaven National Laboratory, Long Island, NY,** *Aerosol-cloud interactions in marine warm clouds and implications for Marine Cloud Brightening.*

Dec. 2022 — **2022 AGU Fall Meeting, Chicago, IL,** *On the Conditionality of Marine Low Cloud Albedo Susceptibility: from Meteorological Conditions to Spatiotemporal Scales.*

Media & Press

- **Phys.org**, *Understanding Earth's hidden east-west symmetry could improve climate models*
- **New Scientist**, *Earth has a mysterious triple symmetry that may influence its climate*
- **CIRES News**, *Clouds shape a newly identified east–west symmetry in Earth's reflected sunlight*
- **Reflective blog post**, *Ensuring continuity for atmospheric research*
- **CIRES Spotlights**, *Flying high in the sky*
- **NOAA CPO News**, *The unintended consequences of reducing sulfur emissions from ships*
- **NOAA Research**, *Scientists turn to artificial intelligence to assess the warming effect of reduced pollution*
- **CIRES News**, *Scientists turn to artificial intelligence to assess the warming effect of reduced pollution*
- **SciTechDaily**, *Cleaner Ships, Hotter Earth: The Unexpected Climate Twist*
- **Mongabay**, *Shipping emissions reduction sheds light on marine cloud geoengineering*
- **DOE ARM NEWS**, *Smoke Studies: Crucial Cloud-Deck Science*

References

Graham Feingold
Senior Science Advisor, Renaissance Philanthropy
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Robert Wood
Professor, University of Washington
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Paquita Zuidema
Professor, University of Miami
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Johannes Mülmenstädt
Earth Scientist, Pacific Northwest National Laboratory
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Publications

Peer-reviewed

21 M. S. Diamond, A. S. Ackerman, A. Baró Pérez, et al., "Drizzle drives model differences in the Southeastern Atlantic Stratocumulus Transitions with Aerosol-Rain-Radiation interactions (SEA STARR) large eddy simulation intercomparison project," *Atmos. Chem. Phys.*, 2026, *under review*.

20 T. Dror, **J. Zhang**, and G. Feingold, "Amazon darkening mitigated by all-sky biophysical feedbacks," *Nature Communications*, 2026, *under review*.

19 **J. Zhang**, J. Gristey, and G. Feingold, "Earth's East-West albedo symmetry," *Nature*, vol. 654, pp. 676–682, 2026. DOI: 10.1038/s41586-026-10624-2

18 **J. Zhang**, D. Painemal, T. Dror, J.-S. Lim, A. Sorooshian, and G. Feingold, "Inferring processes governing cloud transition during mid-latitude marine cold-air outbreaks from satellite," *Atmos. Chem. Phys.*, vol. 26, no. 9, pp. 6015–6034, 2026. DOI: 10.5194/acp-26-6015-2026

17 G. Feingold, F. Glassmeier, **J. Zhang**, and F. Hoffmann, "Opinion: Inferring process from snapshots of cloud systems," *Atmos. Chem. Phys.*, vol. 25, no. 18, pp. 10 869–10 885, 2025. DOI: 10.5194/acp-25-10869-2025

16 **J. Zhang**, Y.-S. Chen, E. Gryspeerdt, T. Yamaguchi, and G. Feingold, "Radiative forcing from the 2020 shipping fuel regulation is large but hard to detect," *Commun. Earth Environ.*, vol. 6, no. 18, pp. 1–11, 2025. DOI: 10.1038/s43247-024-01911-9

15 Y.-S. Chen, **J. Zhang**, F. Hoffmann, T. Yamaguchi, F. Glassmeier, X. Zhou, and G. Feingold, "Diurnal evolution of non-precipitating marine stratocumuli in a large-eddy simulation ensemble," *Atmos. Chem. Phys.*, vol. 24, no. 22, pp. 12 661–12 685, 2024. DOI: 10.5194/acp-24-12661-2024

14 G. Feingold, V. Ghate, L. M. Russell, et al., "Physical science research needed to evaluate the viability and risks of marine cloud brightening," *Science Advances*, vol. 10, no. 12, eadi8594, 2024. DOI: 10.1126/sciadv.adi8594

13 **J. Zhang**, Y.-S. Chen, T. Yamaguchi, and G. Feingold, "Cloud water adjustments to aerosol perturbations are buffered by solar heating in non-precipitating marine stratocumuli," *Atmos. Chem. Phys.*, vol. 24, no. 18, pp. 10 425–10 440, 2024. DOI: 10.5194/acp-24-10425-2024

12 C. Howes, P. E. Saide, H. Coe, et al., "Biomass-burning smoke's properties and its interactions with marine stratocumulus clouds in WRF-CAM5 and southeastern Atlantic field campaigns," *Atmos. Chem. Phys.*, vol. 23, no. 21, pp. 13 911–13 940, 2023. DOI: 10.5194/acp-23-13911-2023

11 **J. Zhang** and G. Feingold, "Distinct regional meteorological influences on low-cloud albedo susceptibility over global marine stratocumulus regions," *Atmos. Chem. Phys.*, vol. 23, no. 2, pp. 1073–1090, 2023. DOI: 10.5194/acp-23-1073-2023

10 P. A. Barrett, S. J. Abel, H. Coe, et al., "Intercomparison of airborne and surface-based measurements during the clarify, oracles and lasic field experiments," *Atmos. Meas. Tech.*, vol. 15, no. 21, pp. 6329–6371, 2022. DOI: 10.5194/amt-15-6329-2022

9 M. S. Diamond, P. E. Saide, P. Zuidema, et al., "Cloud adjustments from large-scale smoke–circulation interactions strongly modulate the southeastern atlantic stratocumulus-to-cumulus transition," *Atmos. Chem. Phys.*, vol. 22, no. 18, pp. 12 113–12 151, 2022, (ACP highlight). DOI: 10.5194/acp-22-12113-2022

8 **J. Zhang**, X. Zhou, T. Goren, and G. Feingold, "Albedo susceptibility of northeastern pacific stratocumulus: The role of covarying meteorological conditions," *Atmos. Chem. Phys.*, vol. 22, no. 2, pp. 861–880, 2022. DOI: 10.5194/acp-22-861-2022

7 **J. Zhang** and P. Zuidema, "Sunlight-absorbing aerosol amplifies the seasonal cycle in low-cloud fraction over the southeast atlantic," *Atmos. Chem. Phys.*, vol. 21, no. 14, pp. 11 179–11 199, 2021. DOI: 10.5194/acp-21-11179-2021

6. X. Zhou, **J. Zhang**, and G. Feingold, "On the importance of sea surface temperature for aerosol-induced brightening of marine clouds and implications for cloud feedback in a future warmer climate," *Geophys. Res. Lett.*, vol. 48, no. 24, e2021GL095896, 2021. DOI: https://doi.org/10.1029/2021GL095896

5. S. J. Abel, P. A. Barrett, P. Zuidema, **J. Zhang**, M. Christensen, F. Peers, J. W. Taylor, I. Crawford, K. N. Bower, and M. Flynn, "Open cells exhibit weaker entrainment of free-tropospheric biomass burning aerosol into the south-east Atlantic boundary layer," *Atmos. Chem. Phys.*, vol. 20, no. 7, pp. 4059–4084, 2020. DOI: 10.5194/acp-20-4059-2020

4. **J. Zhang** and P. Zuidema, "The diurnal cycle of the smoky marine boundary layer observed during August in the remote southeast Atlantic," *Atmos. Chem. Phys.*, vol. 19, no. 23, pp. 14 493–14 516, 2019, (ACP highlight). DOI: 10.5194/acp-19-14493-2019

3. A. S. Chandra, P. Zuidema, S. Krueger, A. Kochanski, S. P. de Szoeke, and **J. Zhang**, "Moisture distributions in tropical cold pools from equatorial Indian ocean observations and cloud-resolving simulations," *J. Geophys. Res. Atmos.*, vol. 123, no. 20, pp. 11, 445–11, 465, 2018. DOI: 10.1029/2018JD028634

2. **J. Zhang**, P. Zuidema, D. D. Turner, and M. P. Cadeddu, "Surface-based microwave humidity retrievals over the equatorial Indian ocean: Applications and challenges," *J. Appl. Meteor. Climatol.*, vol. 57, no. 8, pp. 1765–1782, 2018. DOI: 10.1175/JAMC-D-17-0301.1

1. P. Zuidema, A. J. Sedlacek III, C. Flynn, S. Springston, R. Delgadillo, **J. Zhang**, A. C. Aiken, A. Koontz, and P. Muradyan, "The Ascension island boundary layer in the remote southeast Atlantic is often smoky," *Geophys. Res. Lett.*, vol. 45, no. 9, pp. 4456–4465, 2018. DOI: 10.1002/2017GL076926

Other Publications

1. **J. Zhang** and G. Feingold, "Physical Science of Marine Cloud Brightening: Knowledge and Gaps," in *Topical Group on the Physics of Climate*, (invited article), American Physical Society, October 2024, pp. 1–4. URL: https://engage.aps.org/gpc/resources/newsletters